

Cesium WebGL Virtual Globe

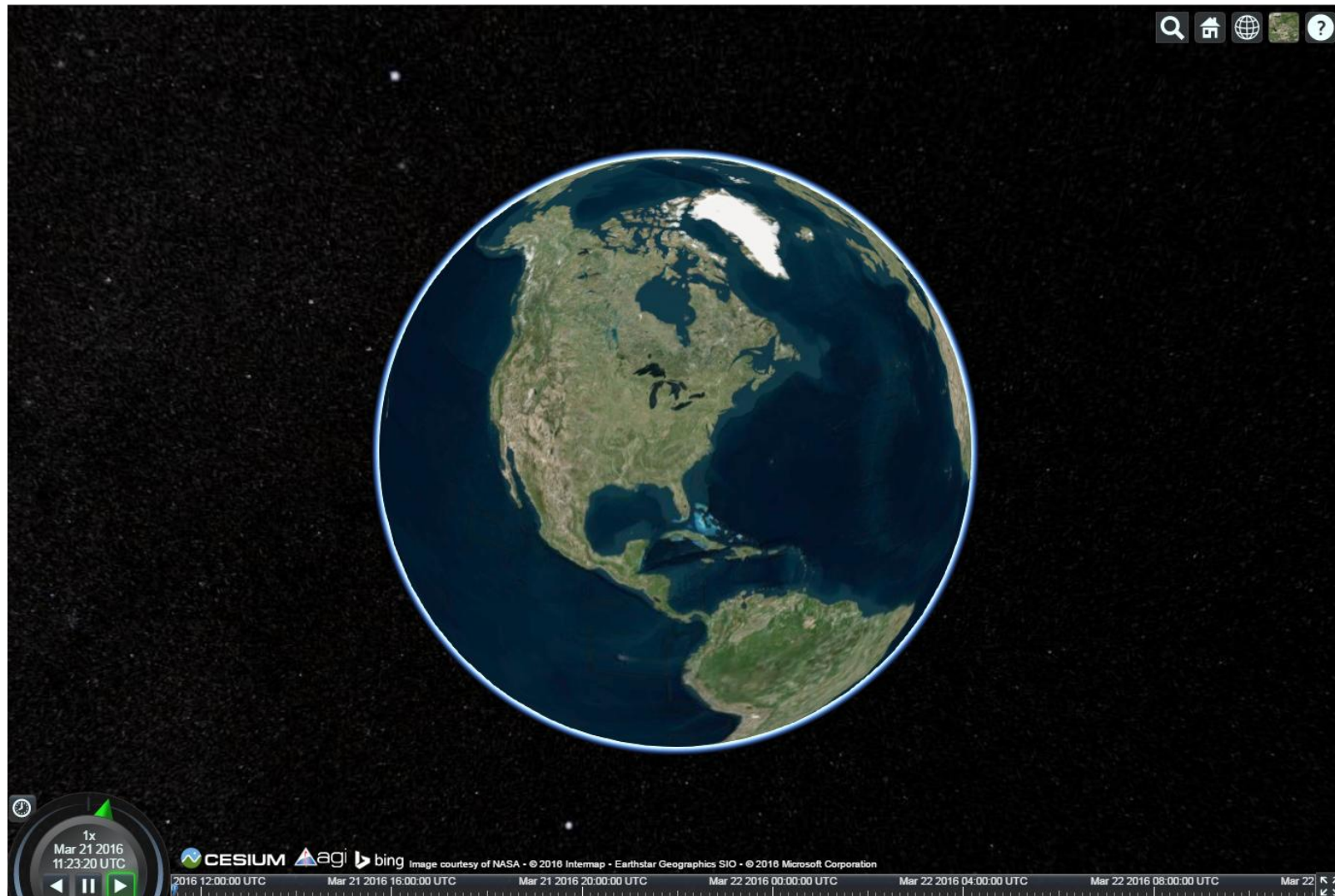
Prof. Dr. Zhihang Yao

Was ist Cesium(JS) WebGL Virtual Globe?

- JavaScript-Bibliothek für 3D-Globen und 2D-Karten
- Auf Basis der Standards WebGL und HTML5 entwickelt
- Läuft nahezu überall (auch auf mobilen Geräten)
- Apache-2.0-Lizenz
- Optimierte für zeitdynamische Daten
- CesiumJS entstand 2011 als Projekt beim US-Unternehmen AGI (Analytical Graphics, Inc.), wurde später als eigenständiges Unternehmen ausgegründet, von einer großen Open-Source-Community unterstützt und gehört seit September 2024 zu Bentley Systems, Incorporated.

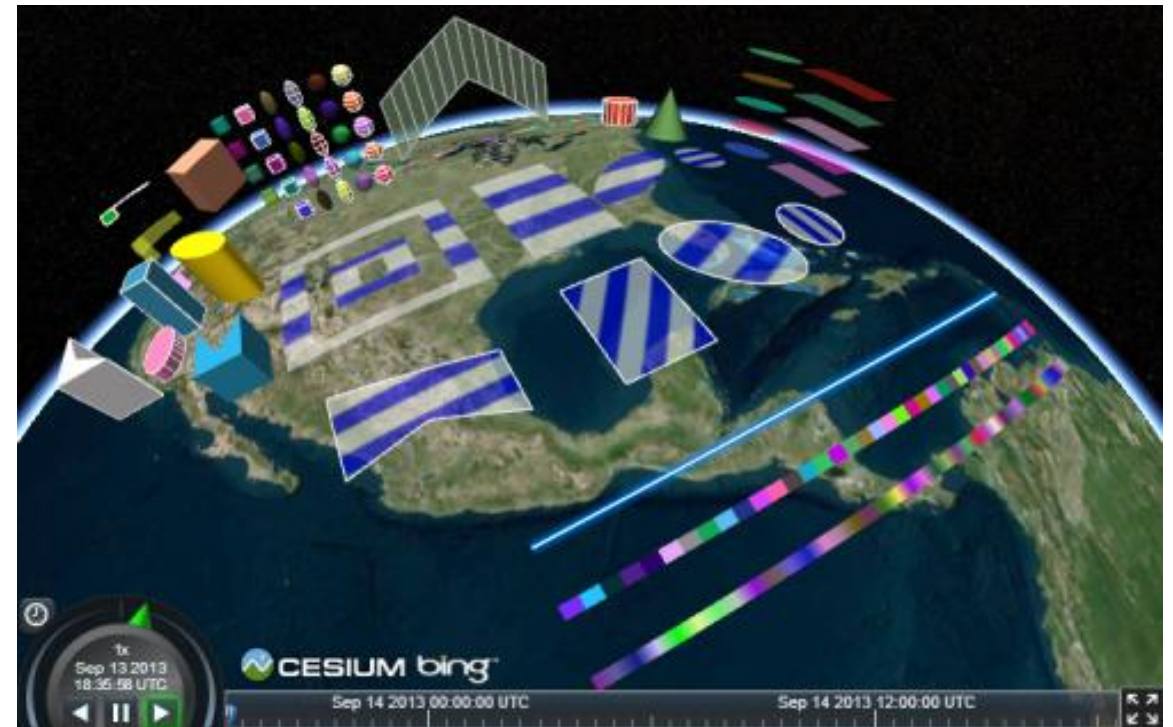


Cesium Viewer



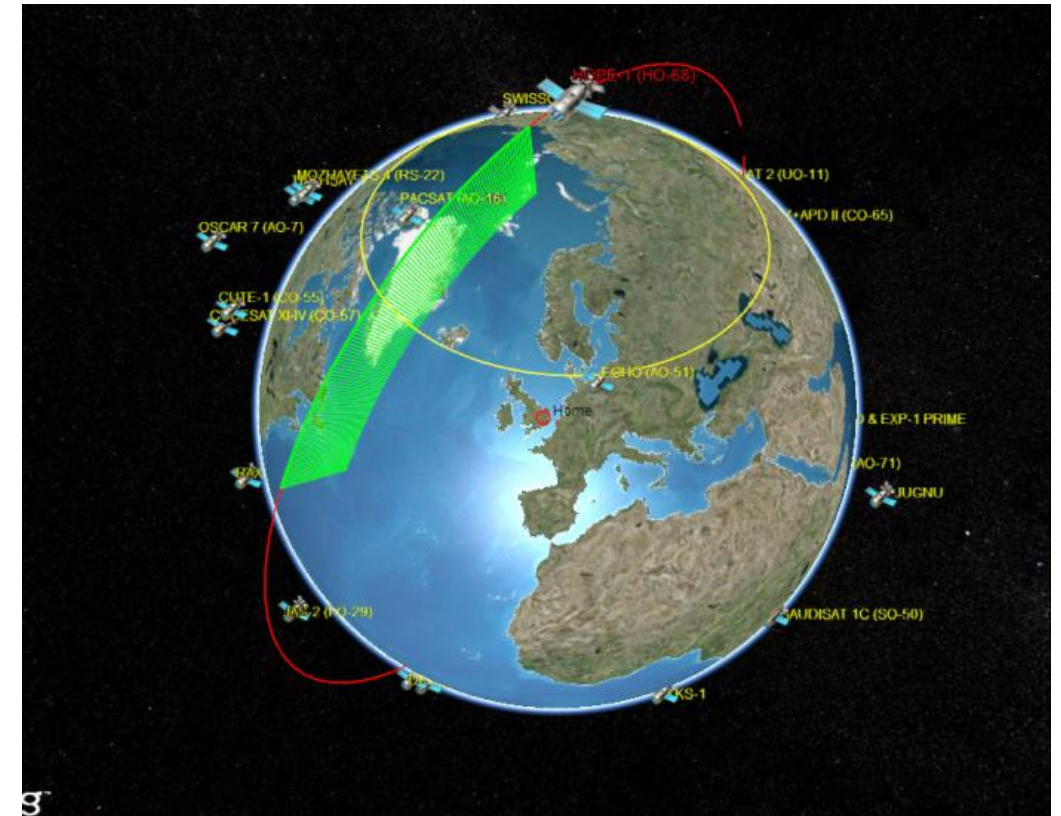
Vektorgeometrie

- Direkte Unterstützung für
 - KML
 - GeoJSON
 - TopoJSON
- Die Cesium-API stellt eine Vielzahl von Objekten und Geometrien dar



Zeitdynamische Visualisierung

- Hochpräzise zeitdynamische Simulation
- Echtzeit-Telemetrie-Streaming
- 4D-Visualisierung
- CZML

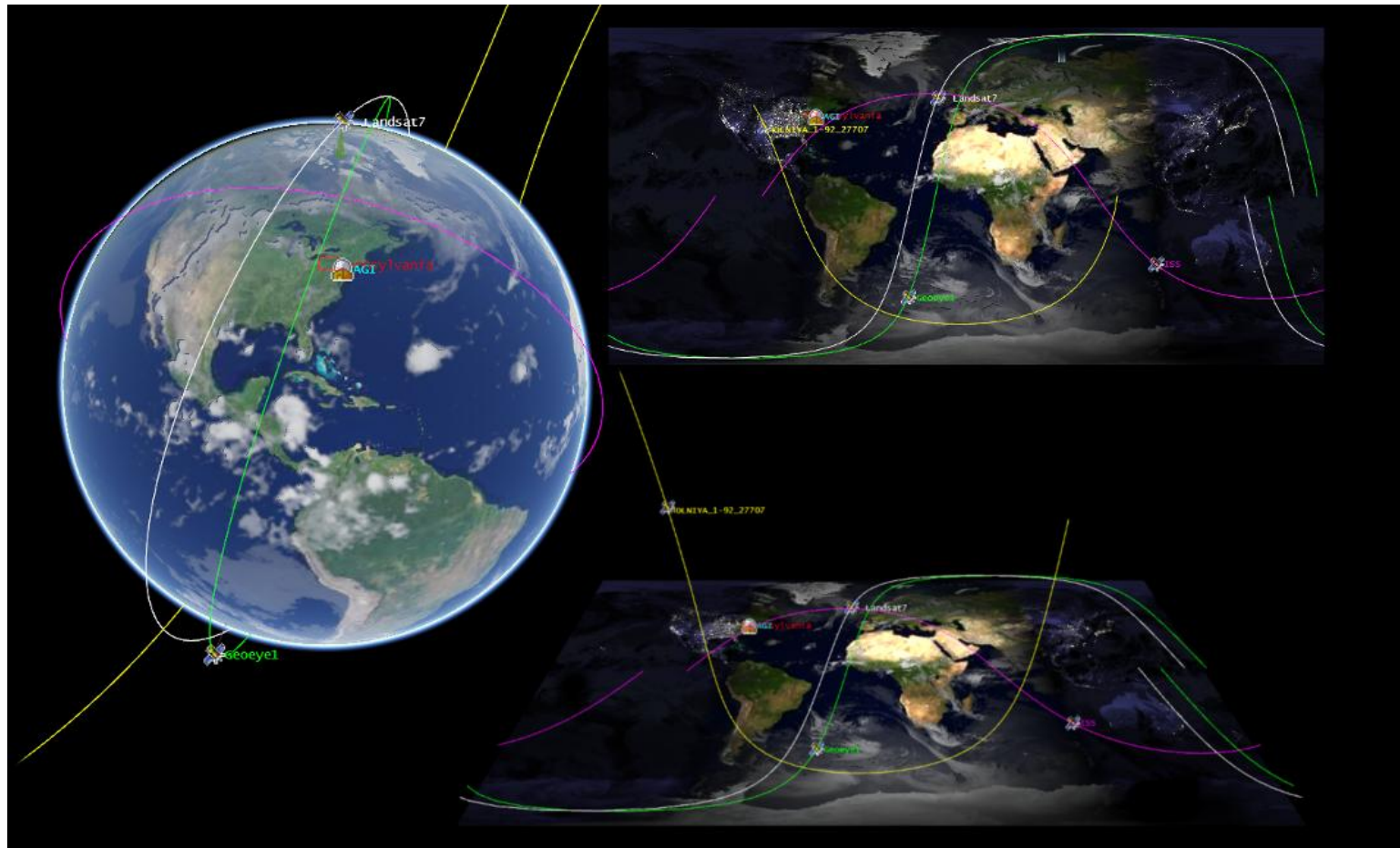


CZML

- Kurzform für Cesium Modeling Language
- Eine streamfähige JSON-Szenenbeschreibung für datengetriebene Visualisierung
- Beschreibt Eigenschaften, die sich im Zeitverlauf ändern
- Unterstützt Zeitvariation für
 - Linien
 - Punkte
 - Modelle
 - Grafische Primitive
- Weitere Details:
 - <https://github.com/AnalyticalGraphicsInc/czml-writer/wiki/CZML-Guide>

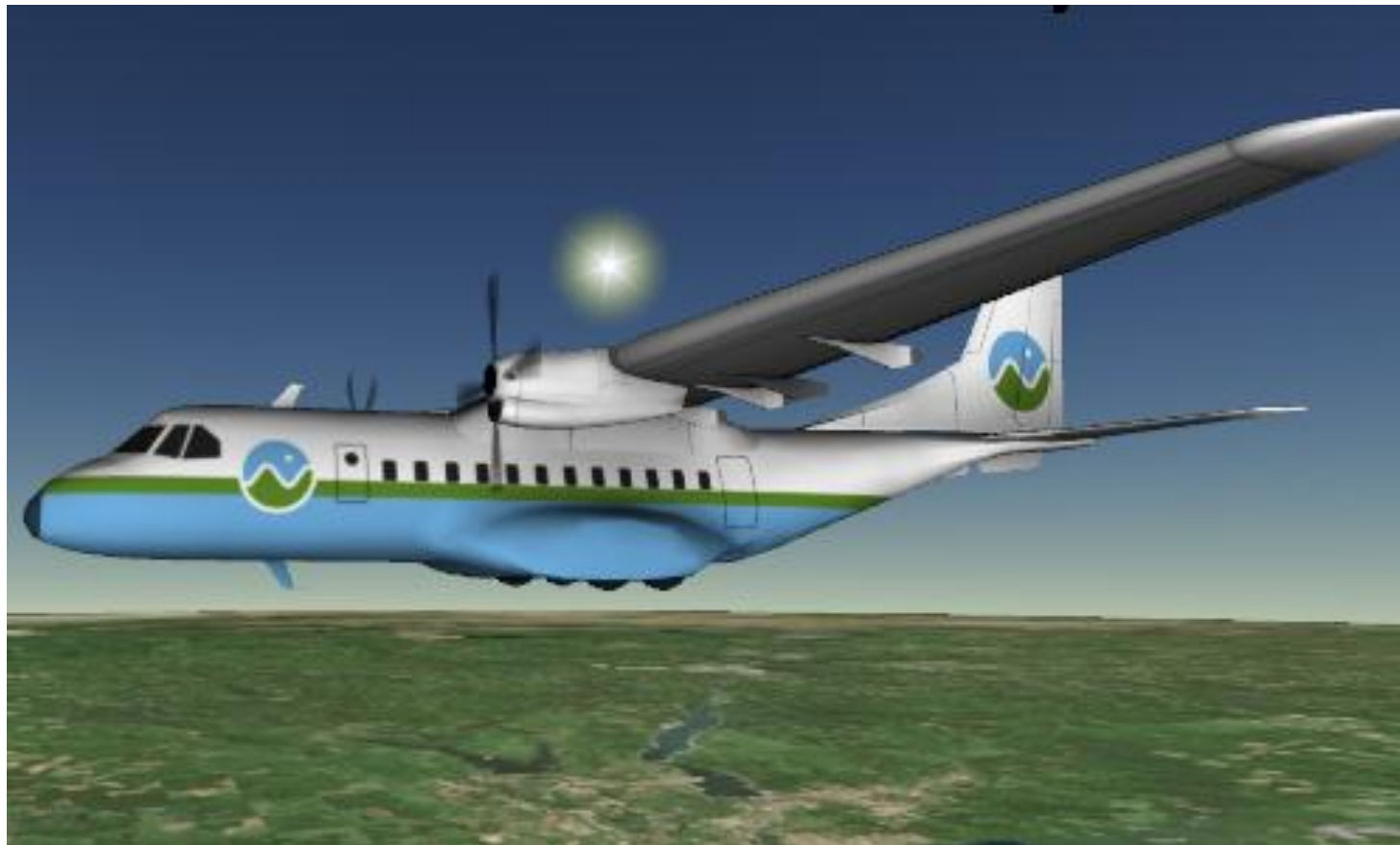
3D-, 2D- und 2,5D-Columbus-Ansicht

- Eine API für drei Ansichten verwenden

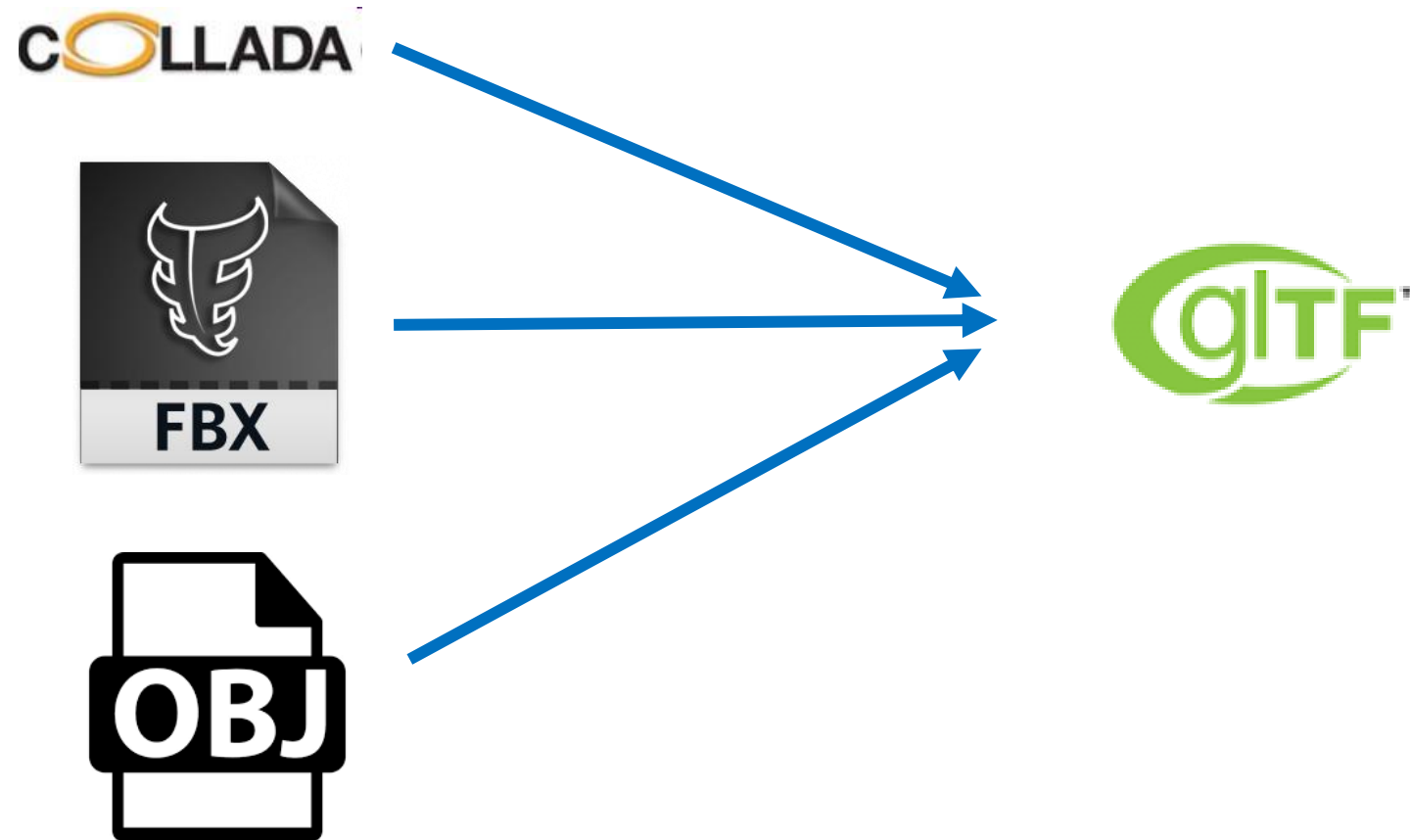


3D-Modelle

- 3D-Modelle mit glTF visualisieren

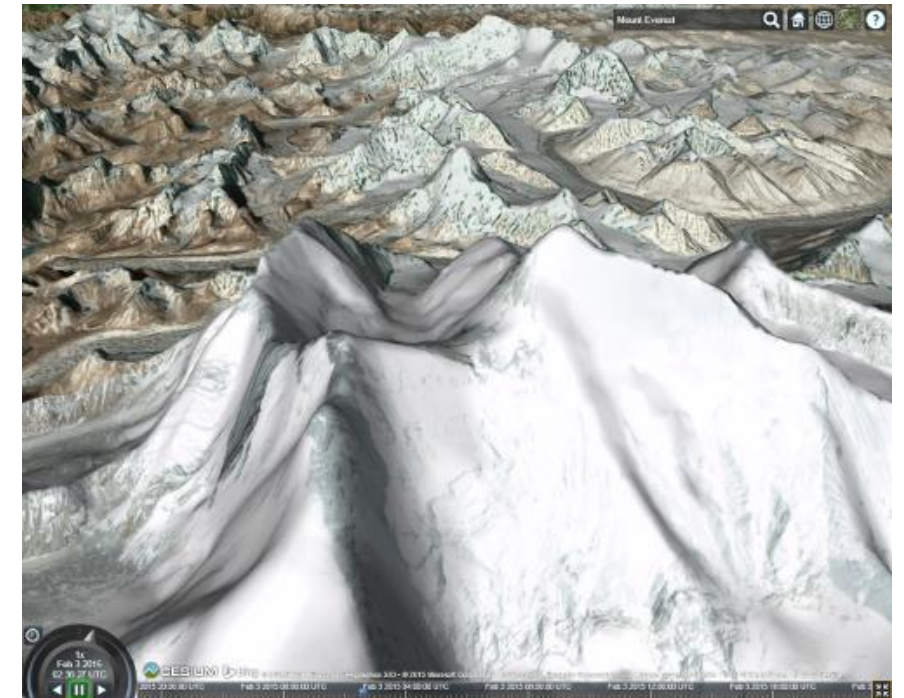


glTF-Konvertierung



Globales Gelände

- Streamen hochaufgelöster globaler Geländedaten, unter Verwendung von
 - Open-Quantized-Mesh-Format
 - Klassische Höhenkarten
- Cesium bietet kostenlosen Zugriff auf SRTM-DEM mit 30 m Auflösung
- Benutzerdefinierte DEMs und DSMs sind möglich

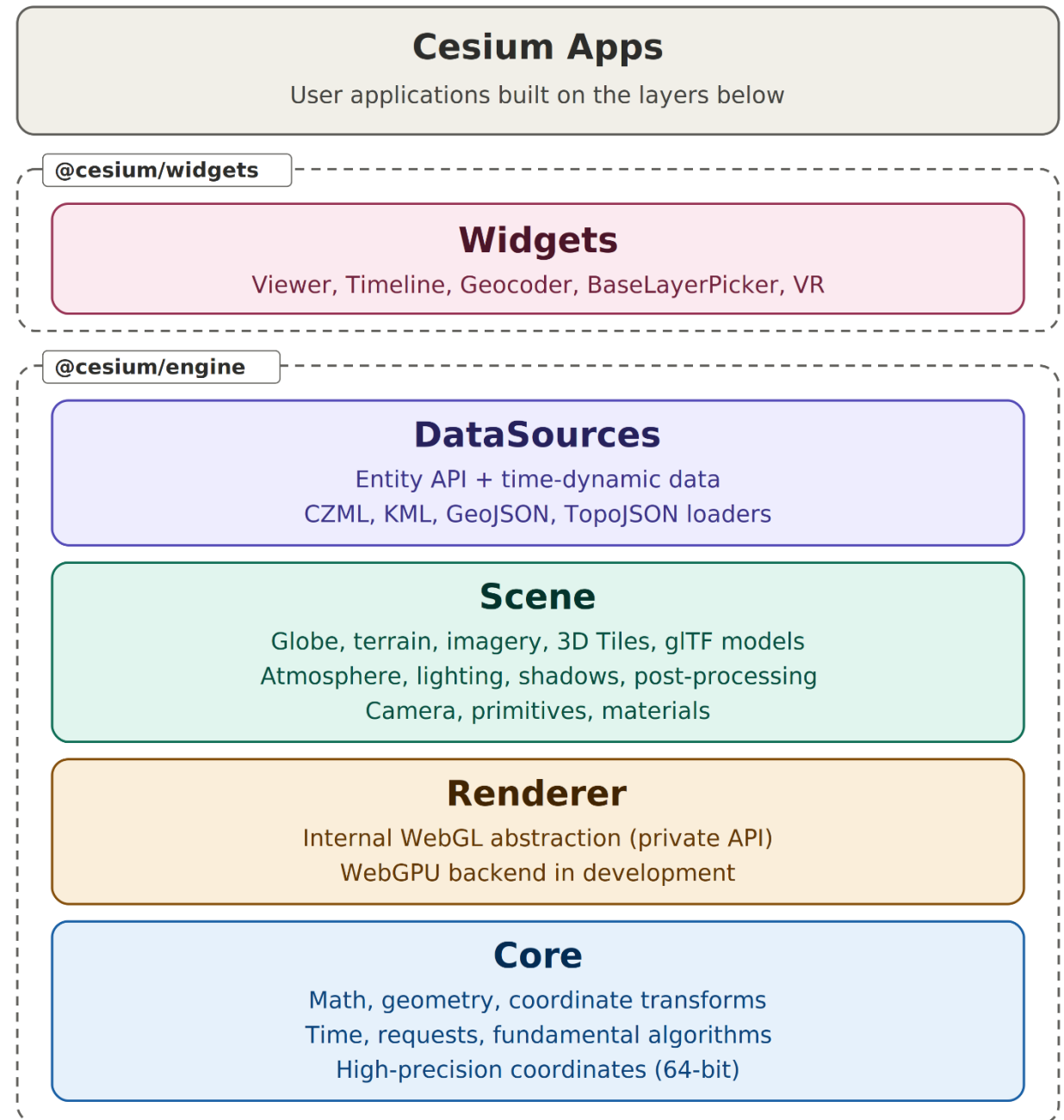


Bildebene (Imagery Layer)

- Streamen von Bildmaterial
 - Unterstützt eine Vielzahl offener Standards
 - Benutzerdefinierte Kachelschemata



Cesium-Architektur

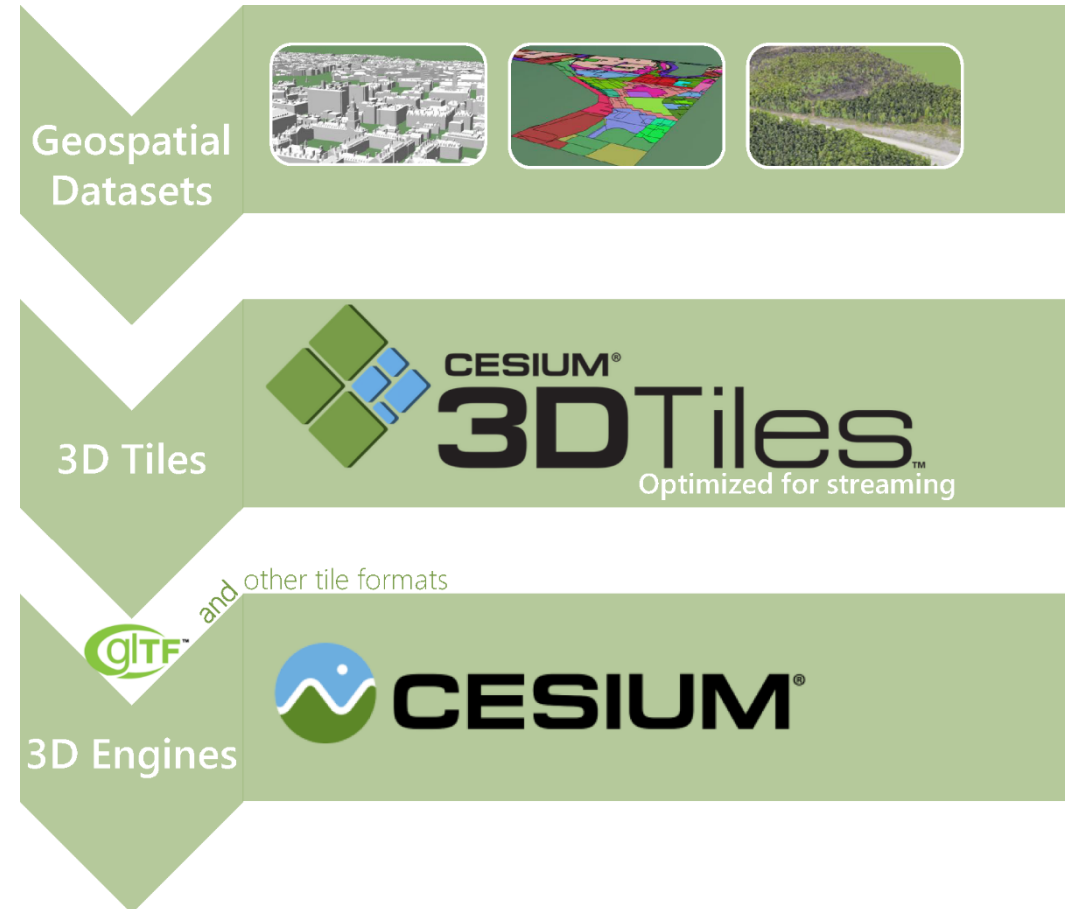


Die fünf Runtime-Engines von Cesium

Engine	Target Platform	Language	Typical Use Cases	License
CesiumJS The original web engine Open-sourced in 2011	Browser / Web WebGL-based	JavaScript npm: @cesium/engine	Web GIS, online maps Browser-based digital twins Trajectory playback, 4D viz	Apache 2.0
Cesium for Unreal Unreal Engine plugin Released in 2021	Unreal Engine 5 Nanite / Lumen support	C++ Built on Cesium Native	Cinematic city visualization Aerospace / defense simulation Real-time ray-traced scenes	Apache 2.0
Cesium for Unity Unity plugin Released in 2022	Unity Cross-platform / mobile / XR	C# + C++ Bindings to Cesium Native	AR / VR / MR applications Mobile 3D mapping Educational and gamified apps	Apache 2.0
Cesium for O3DE Open 3D Engine integration Released in 2022	O3DE (Linux Foundation) Fully open-source engine	C++ Built on Cesium Native	Fully open-source projects Robotics / autonomous-vehicle sim Government / academic research	Apache 2.0
Cesium for Omniverse NVIDIA Omniverse extension Released in 2023	NVIDIA Omniverse Isaac Sim, USD Composer OpenUSD-based	Python + C++ Python for UI Cesium Native for streaming	Industrial digital twins Robotics / AI sim (Isaac) RTX ray-traced large scenes	Apache 2.0

3D Tiles von Cesium

- Eine offene Spezifikation zum Streamen umfangreicher 3D-Geodatensätze
 - Gebäude
 - Bäume
 - Punktwolken
 - Zeitliche Daten
- Verwendet glTF für Modelle



Cesium Sandcastle – Live-Coding

- Geeignet für Einsteiger
 - Einfaches Programmieren in JavaScript
 - Ergebnisse sind sofort sichtbar
- Sandcastle ist auf dem Cesium-Server verfügbar
 - Keine Installation erforderlich

Cesium Sandcastle

The screenshot displays the Cesium Sandcastle web application. The interface is divided into a left sidebar and a main content area.

Left Sidebar:

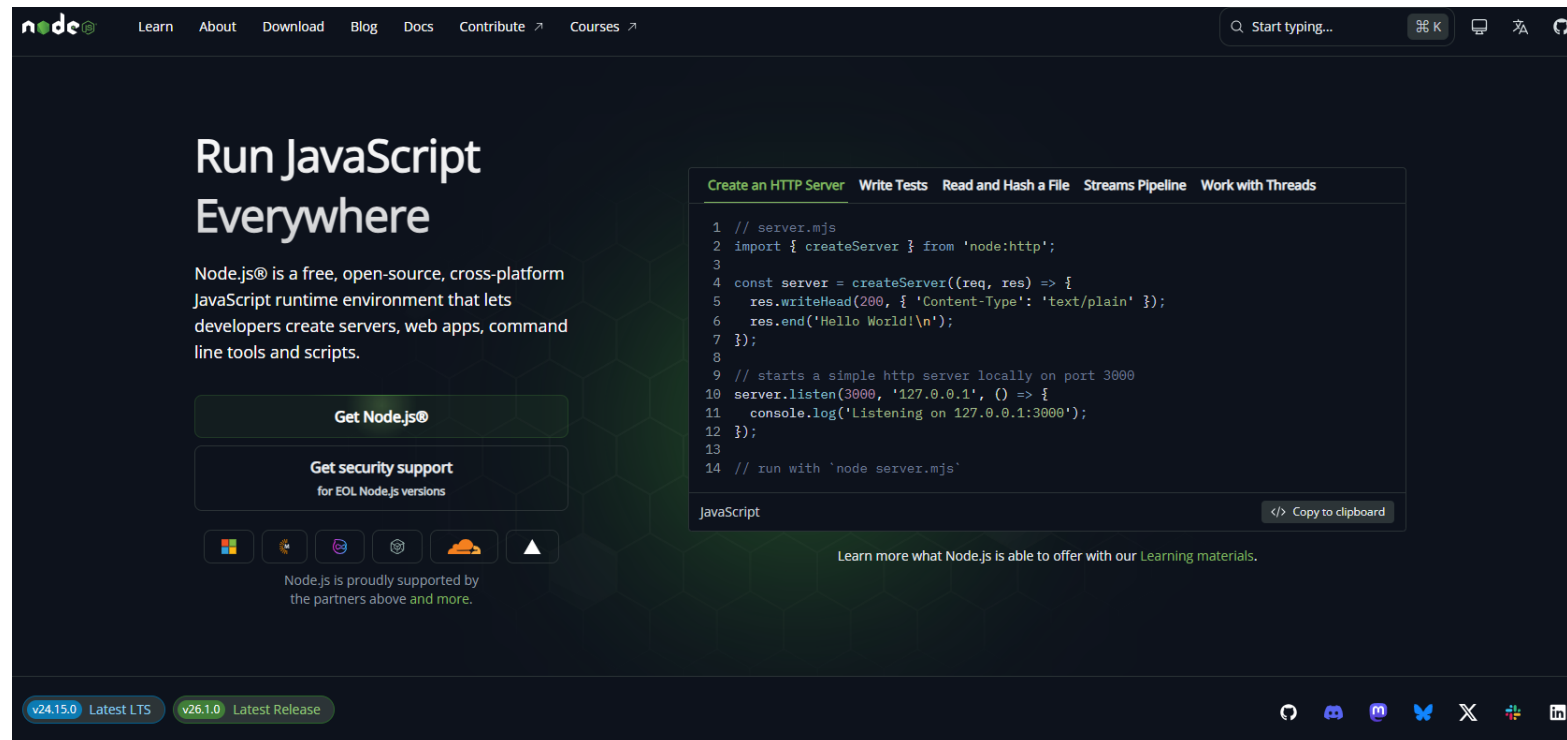
- Gallery:** A section showing search results for "Showcases" (22 results out of 223 total). It lists four showcases with thumbnails and descriptions:
 - 3D Tiles Gaussian Splats – Multiple LODs:** A Gaussian Splats 3D Tiles dataset using multiple levels of detail (LOD). As you zoom in, the dataset transitions from a lower LOD with fewer, larger splats to a higher LOD with more detailed splats, balancing detail, visual fidelity, and performance. (44 lines)
 - 3D Tiles Gaussian Splats:** A minimal code example of a 3D Gaussian splatting (or Gaussian splats) dataset rendered with 3D Tiles. This technique provides smooth, photorealistic surfaces, and is typically derived from reality capture, LIDAR, or photogrammetry data. (19 lines)
 - 3D Tiles Gaussian Splats – Mesh Comparison:** Compare a 3D Tiles power station dataset with Gaussian splatting (or Gaussian splats) rendering versus the same dataset using triangle-based mesh rendering. Notably, fine details like powerlines are better preserved when rendering the Gaussian splatting dataset. (65 lines)
 - Vertical Exaggeration with 3D Tiles:** Apply vertical exaggeration globally to Google Photorealistic 3D Tiles. Exaggerating terrain or building heights can make subtle features easier to see—like highlighting elevation differences at scale or emphasizing aspects of design models. (65 lines)
- Design Model – Layer Control and Metadata Picking:** Explore an architectural design model using layer visibility toggles and metadata-driven picking. Useful for understanding complex AEC datasets, highlighting specific building systems (e.g., facade, HVAC, structural), and

Main Content Area:

- Map:** A large, high-resolution globe view of the Earth, centered on North America.
- Navigation:** A compass and a timeline at the bottom left of the map area. The timeline shows the current time as May 19, 2026, 17:30:28 UTC.
- Footer:** The Cesium logo and version number (1.141.0) are visible in the bottom right corner.

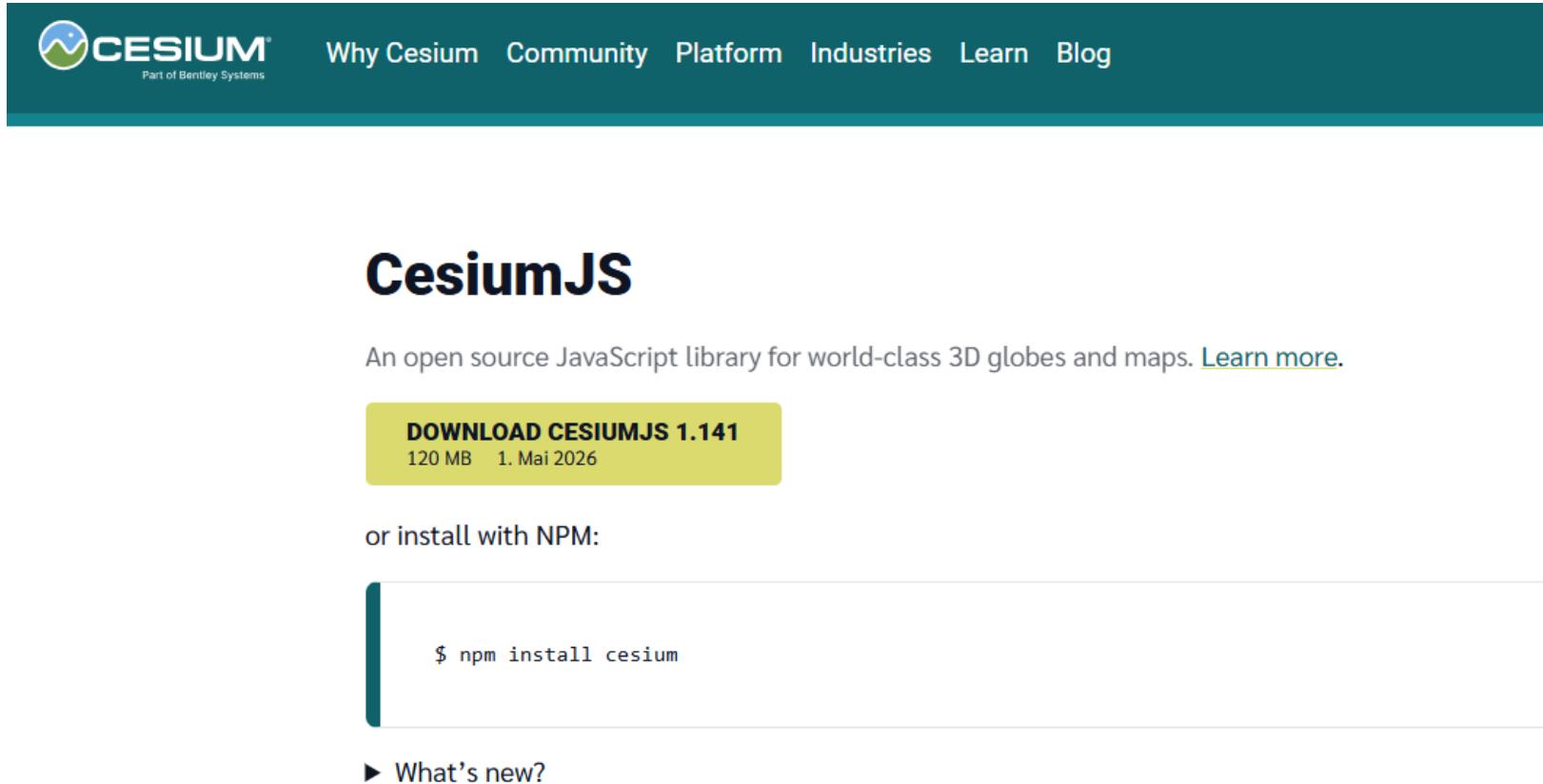
Cesium installieren

- Für die eigene Anwendungsentwicklung wird eine lokale Installation von Cesium empfohlen
- Schritt 1: node.js installieren <https://nodejs.org/>



Cesium installieren

- Schritt 2: Aktuelle Version von Cesium herunterladen
<https://cesium.com/downloads/>



The screenshot shows the CesiumJS website. At the top is a dark teal navigation bar with the Cesium logo (a globe icon and the text 'CESIUM Part of Bentley Systems') and links for 'Why Cesium', 'Community', 'Platform', 'Industries', 'Learn', and 'Blog'. Below the navigation bar, the heading 'CesiumJS' is displayed in a large, bold, black font. Underneath the heading is a descriptive sentence: 'An open source JavaScript library for world-class 3D globes and maps. [Learn more.](#)'. A prominent yellow button contains the text 'DOWNLOAD CESIUMJS 1.141' and '120 MB 1. Mai 2026'. Below the button, the text 'or install with NPM:' is shown. Underneath this is a code block with a dark teal background and a light blue border, containing the command '\$ npm install cesium'. At the bottom of the page, there is a link '► What's new?'.

CesiumJS

An open source JavaScript library for world-class 3D globes and maps. [Learn more.](#)

DOWNLOAD CESIUMJS 1.141
120 MB 1. Mai 2026

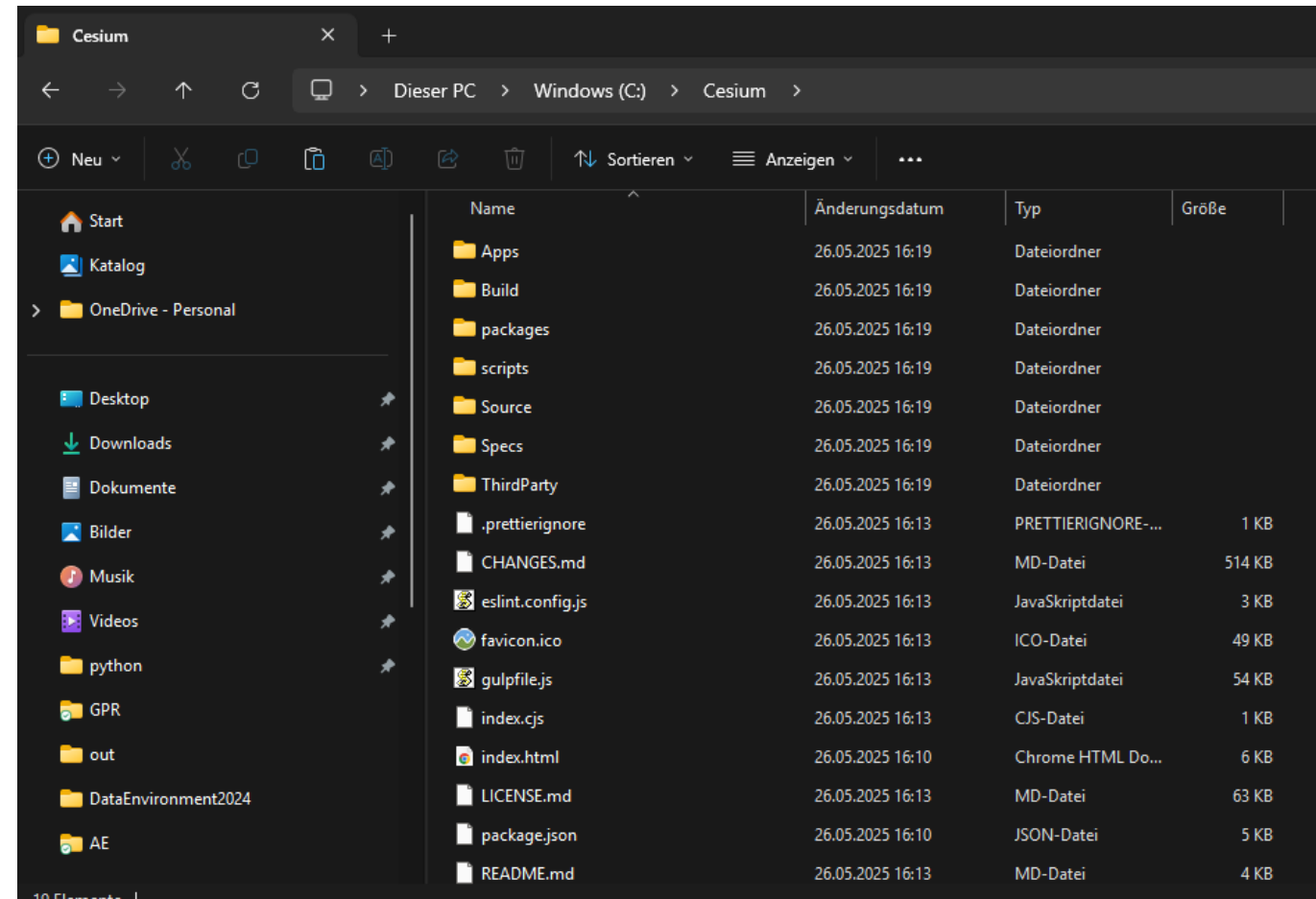
or install with NPM:

```
$ npm install cesium
```

► What's new?

Cesium installieren

- Schritt 3: Den Ordner in einem lokalen Verzeichnis ablegen und entpacken
 - Zum Beispiel: C:\Cesium



Cesium ausführen (unter Windows)

- Schritt 1: Eingabeaufforderung öffnen
- Schritt 2: In das Verzeichnis C:\Cesium wechseln
- Ausführen
 - npm install

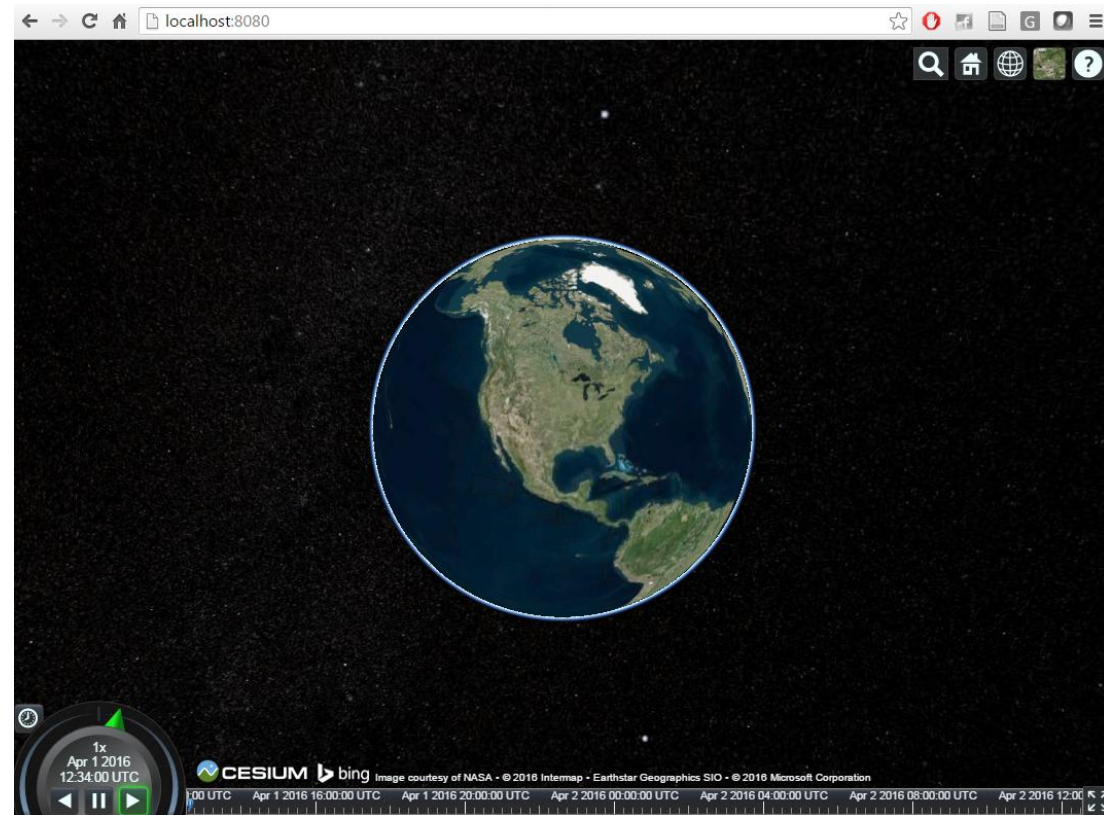
```
C:\Cesium>npm install
```

- node server.js --production

```
C:\Temp\Cesium>node server.js --production  
Sandcastle mirror server running on port 8081  
Cesium development server running locally.  Connect to http://localhost:8080/
```

Cesium ausführen

- Schritt 4: Einen Webbrowser öffnen
 - Zu `http://localhost:8080` wechseln und `HelloWorld.html` auswählen
- Cesium ist einsatzbereit!



Einfaches Hello-World-Beispiel

- HelloWorld.html

```
<!doctype html>
<html lang="en">
  <head>
    <!-- Use correct character set. -->
    <meta charset="utf-8" />
    <!-- Tell IE to use the latest, best version. -->
    <meta http-equiv="X-UA-Compatible" content="IE=edge" />
    <!-- Make the application on mobile take up the full browser screen and disable user scaling. -->
    <meta
      name="viewport"
      content="width=device-width, initial-scale=1, maximum-scale=1, minimum-scale=1, user-scalable=no"
    />
    <title>Hello World!</title>
    <script src="../Build/CesiumUnminified/Cesium.js"></script>
    <style>
      @import url(../Build/CesiumUnminified/Widgets/widgets.css);
      html,
      body,
      #cesiumContainer {
        width: 100%;height: 100%; margin: 0; padding: 0;overflow: hidden;
      }
    </style>
  </head>
  <body>
    <div id="cesiumContainer"></div>
    <script>
      const viewer = new Cesium.Viewer("cesiumContainer");
    </script>
  </body>
</html>
```

```

<!DOCTYPE html>
<html lang="en">
<head>
<title>Adding simple geometry</title>
<script src="../../Build/CesiumUnminified/Cesium.js"></script>
<style>
  @import url(../Build/CesiumUnminified/Widgets/widgets.css);
  html, body, #cesiumContainer {
    width: 100%; height: 100%; margin: 0; padding: 0; overflow: hidden;
  }
</style>
</head>
<body>
<div id="cesiumContainer"></div>
<script>

```

```

var viewer = new Cesium.Viewer('cesiumContainer', { infoBox : false });
var entities = viewer.entities;

```

```

var stripeMaterial = new Cesium.StripeMaterialProperty({
  evenColor : Cesium.Color.WHITE.withAlpha(0.5),
  oddColor : Cesium.Color.BLUE.withAlpha(0.5),
  repeat : 5.0
});

```

```

entities.add({
  rectangle : {
    coordinates : Cesium.Rectangle.fromDegrees(-92.0, 20.0, -86.0, 27.0),
    outline : true,
    outlineColor : Cesium.Color.WHITE,
    outlineWidth : 4,
    stRotation : Cesium.Math.toRadians(45),
    material : stripeMaterial
  }
});
viewer.zoomTo(viewer.entities);

```

```

</script>
</body>
</html>

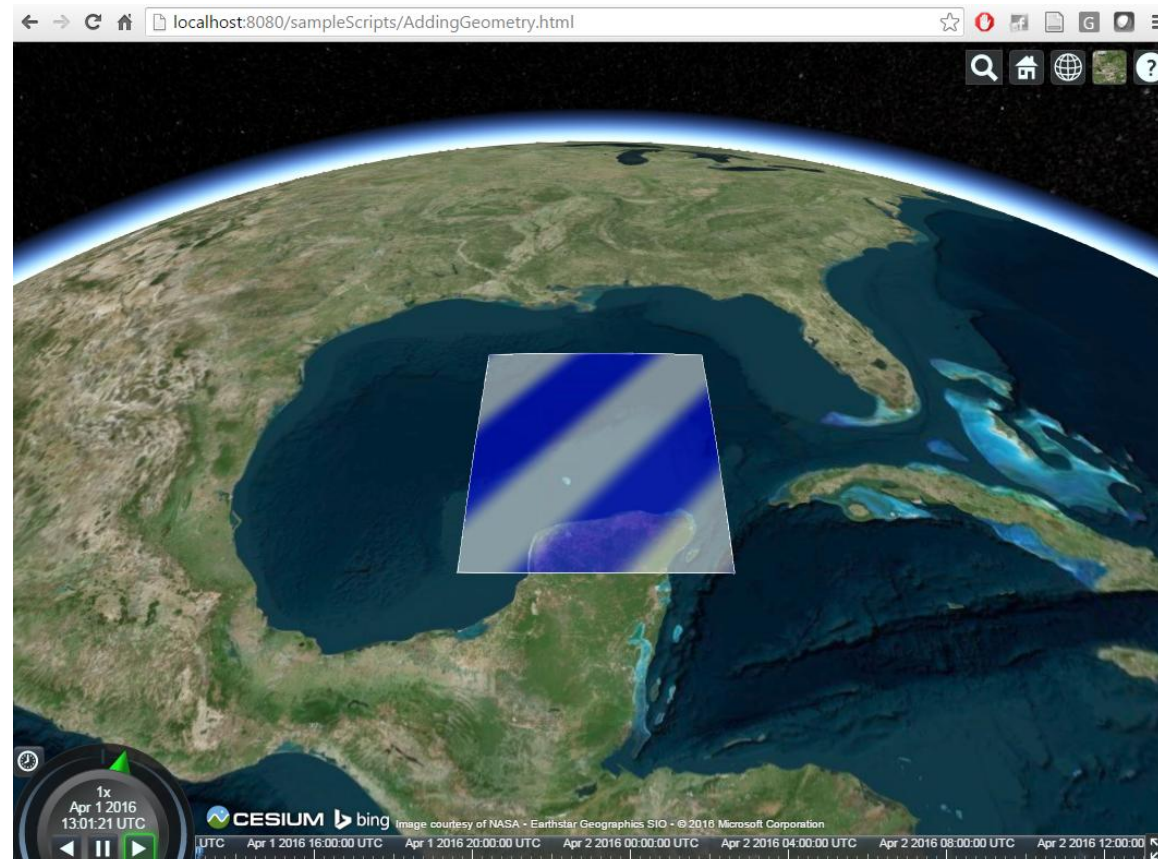
```

Einfache Geometrien hinzufügen

AddingGeometry.html

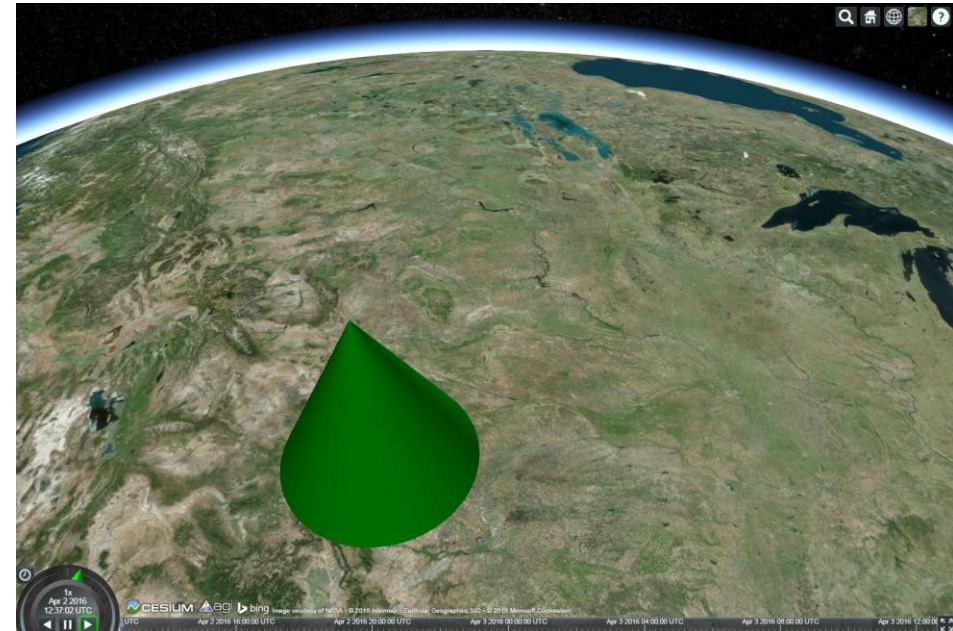
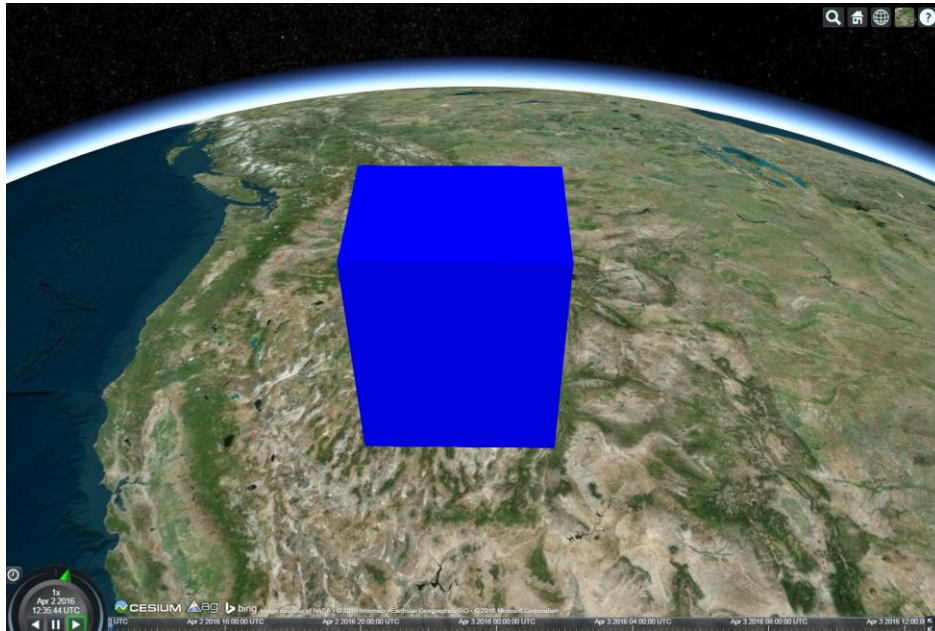
Einfache Geometrien hinzufügen

- AddingGeometry.html im Browser öffnen,
 - z. B. <http://localhost:8080/Apps/AddingGeometry.html> eingeben



Übung

- Erstellen Sie eine neue HTML-Datei „AddingBox.html“ mit einer Box-Geometrie
- Erstellen Sie eine neue HTML-Datei „AddingCone.html“ mit einer Kegel-Geometrie



`<!DOCTYPE html>``.....``<body>``<div id="cesiumContainer"></div>``<script>`

```

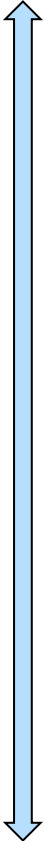
var czml = [
  {
    "id" : "document",
    "name" : "CZML Point - Time Dynamic",
    "version" : "1.0"
  }, {
    "id" : "point",
    "availability" : "2012-08-04T16:00:00Z/2012-08-04T16:05:00Z",
    "position" : {
      "epoch" : "2012-08-04T16:00:00Z",
      "cartographicDegrees" : [
        0, -70, 20, 150000,
        100, -80, 44, 150000,
        200, -90, 18, 150000,
        300, -98, 52, 150000
      ]
    },
    "point" : {
      "color" : {
        "rgba" : [50, 0, 200, 255]
      },
      "outlineColor" : {
        "rgba" : [200, 0, 20, 20]
      },
      "pixelSize" : 15
    }
  }
];

```

```

var viewer = new Cesium.Viewer('cesiumContainer');
viewer.dataSources.add(Cesium.CzmlDataSource.load(czml));

```

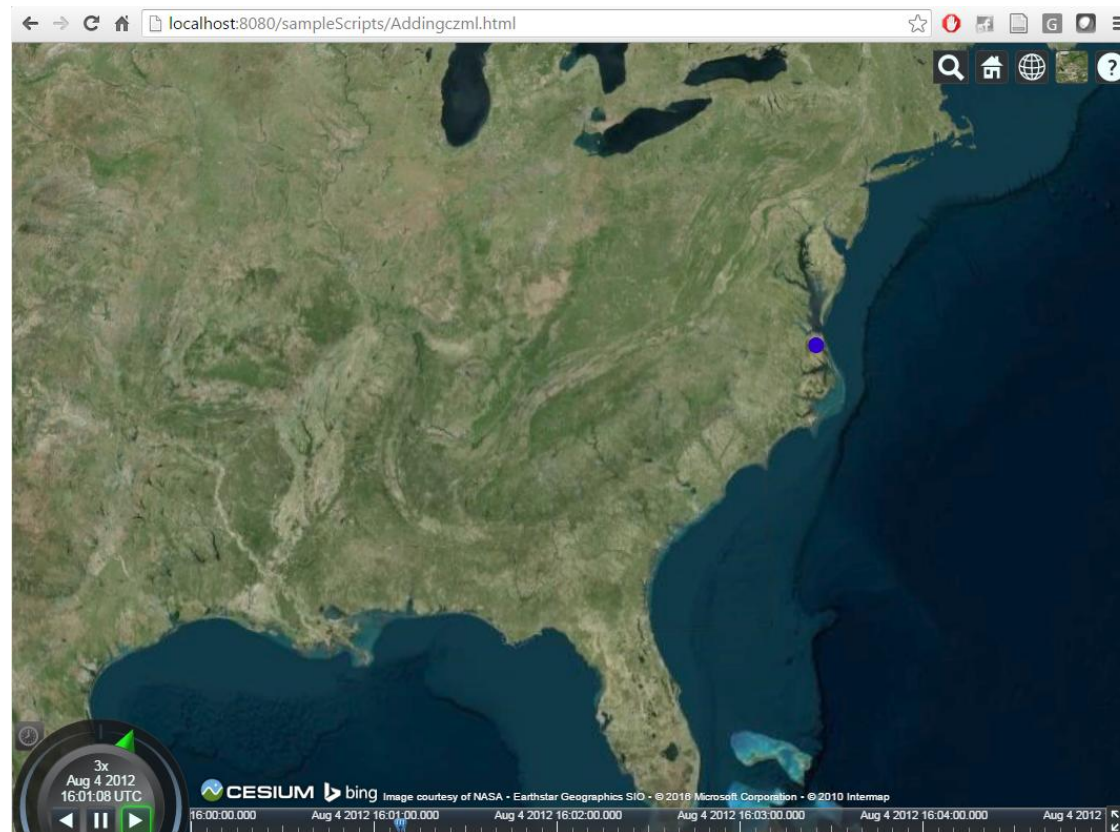
`</script>``</body>``</html>`


CZML definieren

Einfaches CZML
hinzufügen
AddingCZML.html

Einfaches CZML hinzufügen

- AddingCZML.html im Browser öffnen,
 - z. B. <http://localhost:8080/Apps/AddingCZML.html> eingeben



Übung

- Können Sie versuchen, einen weiteren dynamischen Punkt hinzuzufügen, der sich parallel zu diesem Punkt bewegt?

```

<!DOCTYPE html>
<html lang="en">
<head>
<title>Adding KML</title>
<script src="../../Build/CesiumUnminified/Cesium.js"></script>
<style>
  @import url(../Build/CesiumUnminified/Widgets/widgets.css);
  html, body, #cesiumContainer {
    width: 100%; height: 100%; margin: 0; padding: 0; overflow: hidden;
  }
</style>
</head>
<body>
<div id="cesiumContainer"></div>
<script>

```

```

var viewer = new Cesium.Viewer('cesiumContainer', { infoBox : false });
var entities = viewer.entities;

```

```

var options = {
  camera : viewer.scene.camera,
  canvas : viewer.scene.canvas
};

```

```

viewer.dataSources.add(Cesium.KmlDataSource.load('SampleData/kml/bikeRide.kml', options)).then(function(dataSource){
  viewer.clock.shouldAnimate = false;
  var rider = dataSource.entities.getById('tour');
  viewer.flyTo(rider).then(function(){
    viewer.trackedEntity = rider;
    viewer.selectedEntity = viewer.trackedEntity;
    viewer.clock.multiplier = 30;
    viewer.clock.shouldAnimate = true;
  });
});

```

```

</script>
</body>
</html>

```

Datenquelle hinzufügen – KML

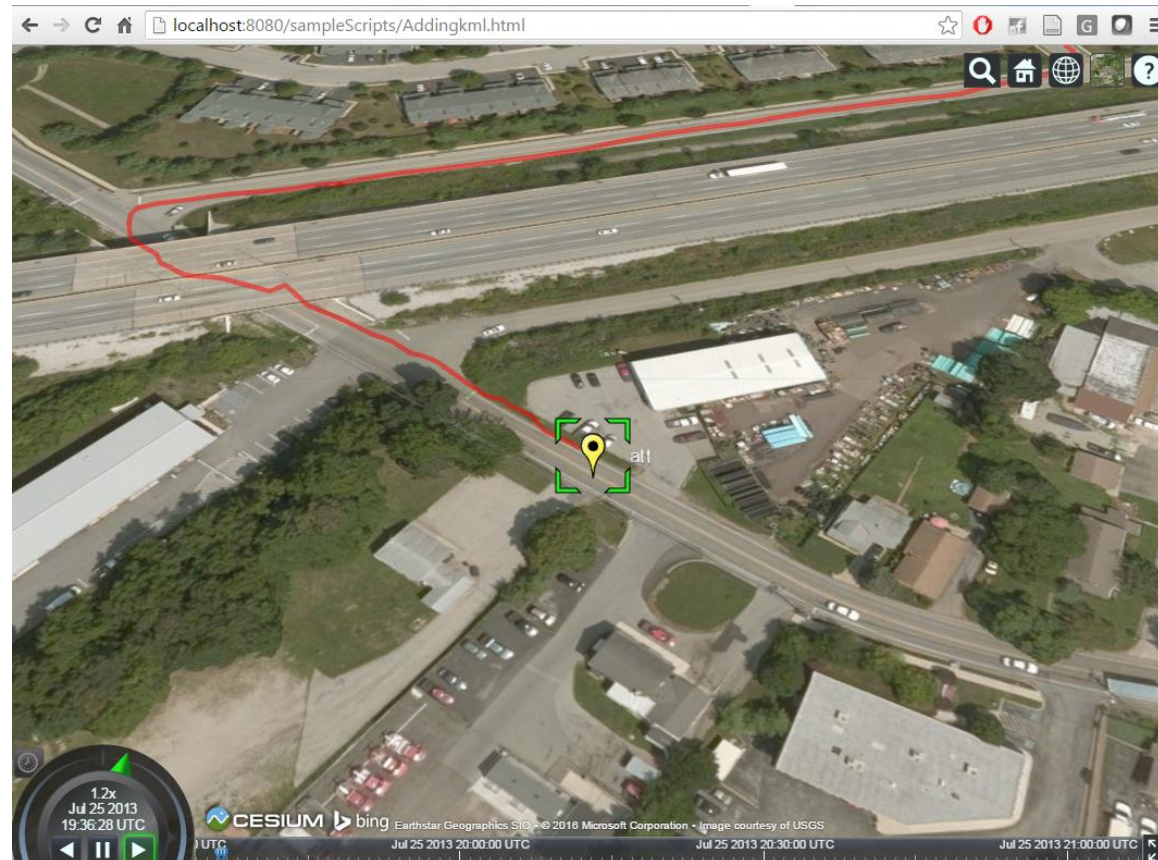
AddingKML.html

KML-Quelle
hinzufügen



Datenquelle hinzufügen – KML

- AddingKML.html im Browser öffnen,
 - z. B. <http://localhost:8080/Apps/AddingKML.html> eingeben



Einfaches glTF-3D-Modell

- Model3D.html

```
<script>
    const viewer = new Cesium.Viewer("cesiumContainer", {
        infoBox: false,
        selectionIndicator: false,
        shadows: true,
        shouldAnimate: true,
    });
    const position = Cesium.Cartesian3.fromDegrees(-123, 44, 150);
    const heading = Cesium.Math.toRadians(135);
    const pitch = 0;
    const roll = 0;
    const hpr = new Cesium.HeadingPitchRoll(heading, pitch, roll);
    const orientation = Cesium.Transforms.headingPitchRollQuaternion(position, hpr);
    const entity = viewer.entities.add({
        name: 'Model',
        position: position,
        orientation: orientation,
        model: {
            uri: 'SampleData/models/CesiumDrone/CesiumDrone.glb',
        },
    });
    viewer.trackedEntity = entity;
</script>
```

Einfaches glTF-3D-Modell

- Model3D.html im Browser öffnen,
 - z. B. <http://localhost:8080/Apps/Model3D.html> eingeben



Übung

- Können Sie versuchen, parallel zu dieser Drohne ein weiteres 3D-Modell hinzuzufügen?